

Autonomic and attentional pathways in the emergence of autism: bridging mechanisms and real-world contexts in infancy

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Abstract

Autism spectrum disorder (ASD) unfolds over the first two years of life through complex interactions among developmental systems. Attention and autonomic nervous system (ANS) regulation represent foundational processes critical for adaptive engagement with the environment. Disruptions in these systems during early infancy may initiate developmental cascades that contribute to core ASD features, including social-communication challenges and restricted and repetitive behaviors, as well as the vast heterogeneity found within ASD. This chapter reviews our approach to integrating mechanistic research and real-world methodologies to explore autonomic and attentional pathways in the early development of ASD.

We conduct prospective, longitudinal studies to track how ANS regulation and attention unfold in infancy in infants at elevated and low familial likelihood for ASD. By integrating multimethod approaches to study these functions in naturalistic contexts, we demonstrate that autonomically regulated attention mechanisms support key developmental milestones and may play an important role in the emergence of ASD. Findings presented provide preliminary support for the theory that infants later diagnosed with ASD exhibit early disruptions in parasympathetic modulation of arousal and attention across responses to social and nonsocial contexts.

This work emphasizes the value of leveraging portable, non-invasive technologies to study early autonomic and attention processes in naturalistic contexts. By using this approach to uncover formative mechanisms underlying ASD, we enhance the translational potential of this research while addressing critical gaps in inclusivity and representation. Understanding autonomic and attentional systems in infancy provides an opportunity to identify cascading developmental divergences and promote adaptive developmental trajectories in ASD.

Keywords: Autism, Autism spectrum disorder (ASD), Infancy, Dynamic systems, Developmental cascades, Respiratory sinus arrhythmia (RSA), Parasympathetic, Autonomic nervous system, Attention, sustained attention, social attention, Head-mounted eye tracking, Naturalistic

Autism spectrum disorder (ASD) emerges from the complex interplay of developmental systems, where early differences cascade across developmental domains to shape growth and adaptation over time. Understanding these dynamic processes in the early development of ASD requires bridging theory with application to address critical questions: How do foundational developmental systems unfold in ASD? How do these early developing systems initiate cascading pathways that shape the emergence of ASD? How can our science most directly translate to actionable early identification and intervention strategies that can be implemented during critical developmental windows?

The developmental emergence of core features of ASD occurs gradually over the first three years of life, with diagnostic certainty often achieved by age 2–3 years. Despite most diagnoses occurring between 2–5 years of

age, robust evidence suggests that neurological processes underlying ASD are already underway shortly after birth (see (Dawson et al., 2023) for a review). Core features of ASD include delays and differences in social communication and interaction, alongside repetitive behaviors and restricted interests (American Psychiatric Association, 2013). Prospective longitudinal designs have been used to track the development of infant siblings of children with ASD, with the goal to identify early markers of ASD and advance early detection and intervention methods (Szatmari et al., 2016). These baby sibling research designs comprise cohorts of elevated likelihood infant siblings of children with ASD (EL) and low likelihood infants without a family history of ASD (LL). Approximately 20% of ASD siblings are diagnosed with ASD at age 2–3 years and an additional 30% exhibit atypical development and/or subclinical features of ASD (Ozonoff et al., 2011; Ozonoff et al., 2024). These research designs have been instrumental in advancing the field and bringing us closer to identifying mechanisms and pathways in ASD (Messinger et al., 2013; Ozonoff & Iosif, 2019; Szatmari et al., 2016; Zwaigenbaum et al., 2005; Zwaigenbaum et al., 2015). For example, as a result of infant sibling research, we have identified social-communication delays and repetitive behaviors as early as 9–12 months, revealing early iterative processes that shape and are shaped by subsequent skills and developmental outcomes (Bradshaw et al., 2021; Pileggi et al., 2021). Outside of core features of ASD, findings from such studies have reported even earlier differences in foundational developmental processes, like attention and autonomic nervous system (ANS) functions.

Atypical attention is among the earliest of these developmental processes to characterize individuals with ASD and persists as a lifelong feature of ASD (Ness et al., 2019; Murias et al., 2018; Bradshaw et al., 2019; Kaliukhovich et al., 2020; Frazier et al., 2017). Research has demonstrated that atypical attention is not only detectable in early infancy but is also predictive of later developmental outcomes (Bradshaw et al., 2020; Di Giorgio et al., 2016; Jones & Klin, 2013). However, the biological mechanisms underlying these early disruptions remain poorly understood. One promising avenue of investigation lies in another foundational system—the autonomic nervous system. ANS is an early-emerging bioregulatory system essential for regulating neonatal arousal and attention (Brazelton & Nugent, 2011; Feldman, 2006; Porges & Furman, 2011). Disruptions in the ANS's regulation of attention, or “autonomic regulation of attention,” may initiate a series of cascading effects that influence social interaction, learning, and broader development. While general ANS dysfunction has been documented in older children diagnosed

with ASD (Cheng et al., 2020), less research has focused on whether and when such dysfunction contributes to atypical attention in infancy and how early autonomic regulation predicts the emergence of ASD symptoms.

Baby sibling research to date relies heavily on laboratory-based methods that prioritize experimental control. These approaches, while valuable, simplify the complexities of infants' natural environments, which are rich with experiential and adaptive factors critical to development. Naturalistic data collection—capturing behavior or specific neurodevelopmental processes in everyday environments—represents a conceptual shift in our scientific approach, embracing the dynamic, contextual factors that shape development, rather than removing them in lab-based experiments or controlling for them in statistical analyses (Tamis-LeMonda et al., 2024). By observing infants in their lived environments, researchers can access the very influences that drive developmental trajectories. Studying attention and ANS in these contexts, while leveraging portable, noninvasive technologies, not only advances knowledge of developmental mechanisms, but also bridges the gap between research and practice, facilitating earlier detection and intervention strategies that can be seamlessly implemented in real-world settings.

Naturalistic methods also offer a pathway to address longstanding disparities in participant representation by enabling research in a variety of home settings. Participants in baby sibling studies predominantly comprise white, higher-resource families. Descriptive data from the past three decades of autism intervention research show that white participants have the highest rate of participation, comprising 64.8% of sample cohorts (Steinbrenner et al., 2022). This underrepresentation limits understanding of ASD across diverse populations and addressing these disparities is essential, not only for equity, but also to fully capture the heterogeneity of developmental contributing to the phenotypic diversity of ASD.

Building on emerging findings in attention, ANS regulation, and the dynamic interactions among these systems in the early emergence of ASD, we propose a unified framework that integrates existing literature and our findings to better understand how foundational systems in real-world settings initiate cascading effects in the emergence of ASD. First, we explore how cascading developmental processes—specifically, disruptions in attention and autonomic regulation—shape typically developing and ASD trajectories during infancy. Next, we highlight the promise of portable, noninvasive technologies like head-mounted eye tracking (HMET) and synchronized home-based systems to capture these processes in real-world settings. Finally, we address critical challenges in participant inclusion and

representation, underscoring the importance of studying diverse populations to ensure findings are relevant and translatable to broader contexts. By bridging mechanistic research with innovative methodologies and inclusive approaches to research, we aim to contribute to the current understanding of ASD development in infancy and create more direct pathways toward early detection and intervention.



1. Developmental cascades and the emergence of ASD

A developmental cascades framework views development as continuous, cumulative interactions and transactions across various developmental domains, biological systems, and environmental contexts (Masten & Cicchetti, 2010; Oakes & Rakison, 2019). It provides a lens for investigating how lower- or higher-level processes produce cross-domain, reciprocal effects that influence outcomes later in development. Importantly, cascades highlight both specific developmental mechanisms and the dynamic interactions among multiple mechanisms that unfold over time to shape development. Individual experiences contribute to the divergence of developmental trajectories, creating opportunities for individualized input driven by the infant, which, in turn, influences future developmental pathways.

Developmental cascades offer a useful framework for understanding the early development of ASD, comprised of cascading processes, where early differences in foundational systems, including autonomic and attentional systems, set off a ripple effect that can enrich or disrupt downstream developmental pathways (Bradshaw et al., 2022a; Iverson et al., 2023; Karmiloff-Smith, 1998; Thelen & Smith, 1995). Among these foundational systems, attention stands out as a particularly intriguing process that may be altered in individuals with ASD. Variability in early attentional processes can significantly influence later social, cognitive, and communicative development. In ASD, early attentional differences may contribute to the atypical development of social communication abilities (Bradshaw et al., 2022a). Early attentional processes undergo rapid development over the first few months of life (e.g., by 6–10 weeks), driven by a shift from subcortical to cortical control of behavior. Critical elements of early-developing attention processes—such as disengaging from stimuli, shifting focus, and filtering irrelevant information—are foundational for cognitive and behavioral development (Colombo, 2001; Xie, 2017; Conte & Richards, 2021; Hunnius, 2007). Achievements of these early attentional milestones coupled with co-occurring achievements in other domains (e.g., motor and cognitive)

form a critical foundation for continued growth in social, language, and cognitive domains, making early attention a key focus for understanding the mechanisms underlying atypical developmental trajectories, including those observed in ASD.

In ASD, atypical patterns of visual attention represent the result of, and contributor to, influential developmental cascades that begin in infancy. Reduced attention to socially relevant stimuli, such as eyes and faces, as well as more generalized difficulty disengaging from focal stimuli, emerge within the first year of life (Chawarska et al., 2013; Fu et al., 2024; Jones et al., 2016). These subtle variations in attention may not constitute diagnostic criteria for ASD, but have broad implications for social development. For example, reduced visual saliency of the eyes of a face in early infancy could lead to diminished attention to eyes in later infancy, which may reduce the frequency and quality of interactions with caregivers, resulting in missed opportunities for social learning, and may reinforce nonsocial attentional preferences over time, as is observed in ASD (Pierce et al., 2016).

Atypical visual attention is also evident in young EL infants, irrespective of their eventual diagnostic outcome (Di Giorgio et al., 2021; Di Giorgio et al., 2016). While these attentional differences may be transient (Bradshaw et al., 2020), they point to an underlying genetic basis for atypical social attention in ASD. Genetics strongly influence visual attention patterns throughout childhood (Kennedy et al., 2017; Constantino et al., 2017) which further influence information processing and social interactions. Together, these findings highlight the role that biological predispositions for social or nonsocial stimuli play in the developmental cascades that lead to ASD. ASD-specific differences in lower-level attention processes, such as attention orienting and disengagement, may stem from atypical neural and neurochemical development. Disruptions in arousal regulation and visual processing circuits have been proposed as potential early contributors to attentional variability in ASD (Artoni et al., 2020; Hazlett et al., 2017). Arousal regulation is driven by the sympathetic-parasympathetic balance of the autonomic nervous system. The parasympathetic branch of the ANS supports attention and readiness for cognitive engagement, pointing to the ANS as a critical biological factor in the development of attention, attentional control, and visual processing. Cascading effects of early ANS dysfunction are less well studied, but early differences in ANS and neural circuitry could initiate developmental cascades that affect both regulatory behavior and subsequent brain development, which may shape later patterns of social engagement and learning. For instance, difficulties with both arousal modulation and attention shifting may compound, negatively

affecting infants' ability to engage in triadic interactions with objects and caregivers. These interactions require a calm arousal state and effective attention shifting to facilitate entry into a shared space of joint attention and engagement. Such states of joint engagement with caregivers and objects are critical for language acquisition and higher-order cognitive integration (Adamson et al., 2019; Adamson et al., 2009; Thorp, 2006; Conway et al., 2018).

Overall, the developmental cascades model frames development as a dynamic system, where interactions between biological predispositions, developmental processes, and environmental inputs continuously shape unfolding trajectories. Early disruptions in the foundational systems discussed here, including autonomic regulation and visual attention, can cascade to affect social communication, motor development, and broader neurocognitive functions. Mechanistic frameworks for ASD therefore highlight the convergence of multiple subsystems and underscore the need for integrative, multimodal approaches to studying its emergence.



2. The role of autonomic nervous system function as an early regulatory mechanism in ASD

Systems-level biological factors known to distinguish those with vs. without ASD include genetics, brain development, and physiological arousal. Among these biological systems, the ANS plays a central role in regulating behaviors essential for social and cognitive development. Understanding parasympathetic function as a component of ANS provides insight into its contribution to developmental cascades in ASD. There is robust evidence that ANS-driven physiological systems support behavioral regulation, social engagement and interaction, and attention and cognitive processing (Porges & Furman, 2011; Heilman et al., 2007; Porges, 2003; Richards, 2010; Thayer et al., 2009; Whedon et al., 2018). This body of work lays the foundation for examining how developmental *trajectories* of autonomic and attentional mechanisms in infancy shape development, initiate cascades, and contribute to outcomes.

ANS regulation is governed by the balance of sympathetic and parasympathetic activation, with the sympathetic nervous system activating during times of stress (“fight or flight”) and the parasympathetic nervous system activating during times of rest (“rest and digest”). This delicate balance enacts physiological changes in basic functions, such as decreasing heart rate or increasing body temperature to support infants' healthy responsiveness and adaptation to environmental demands.

In early prenatal development, physiological functions are driven primarily by sympathetic influence. Higher cortical processes begin to integrate autonomic control in the late prenatal and early postnatal period leading to increasing influence of the parasympathetic nervous system. As ANS matures, parasympathetic influences support calm states of alertness and sustained attention that are essential and provide countless opportunities for development across domains, including social and language learning. In this way, ANS is considered a “developmental resource” in the neonatal and early infant period that mediates infant–environment interactions.

Physiological measures used to index parasympathetic function include heart rate variability (HRV) and respiratory sinus arrhythmia (RSA). Both HRV and RSA reflect variability in the influence of the parasympathetic system on the heart, with HRV reflecting variability in heart rate and RSA reflecting variability in heart rate at the frequency of respiration. An adaptive, regulated ANS involves continuous changes to heart rate, making increased variability (higher HRV and RSA) indices of parasympathetic influence over the heart. HRV and RSA undergo significant maturation in infancy and early childhood, and, when measured during periods of rest, our work demonstrates that the first six months of life is a period of particularly rapid growth (Fig. 1). Throughout development, higher HRV and RSA at rest reflect an infant’s readiness and capacity to regulate arousal and maintain engagement with their surroundings. During times of stress, RSA decreases due to the suppression of the parasympathetic system to make way for sympathetic influence. RSA suppression supports mobilization of the fight or flight system, which includes heightened arousal and cognitive focus. Evidence indicates that more RSA suppression in response to stress reflects a healthy, responsive neurophysiological system (Bazhenova et al., 2001; Porges et al., 1996). Studies that demonstrate links between RSA suppression and developmental outcomes, for example social interaction and language, strengthen these claims.

To evaluate the link between RSA suppression and positive developmental outcomes, we conducted a longitudinal study of infants at elevated and low likelihood for ASD (Bradshaw & Abney, 2021). RSA was measured in 3–4-month-old infants during a semi-structured infant–caregiver interaction. The structure of the interaction was modeled after the still-face procedure, but infants and parents were not constrained in terms of positioning or movement. The infant was placed in supine on the Table while the caregiver stood in front of them interacting as they normally would at home for 5 minutes. At that time, the caregiver was cued to sit down and ignore

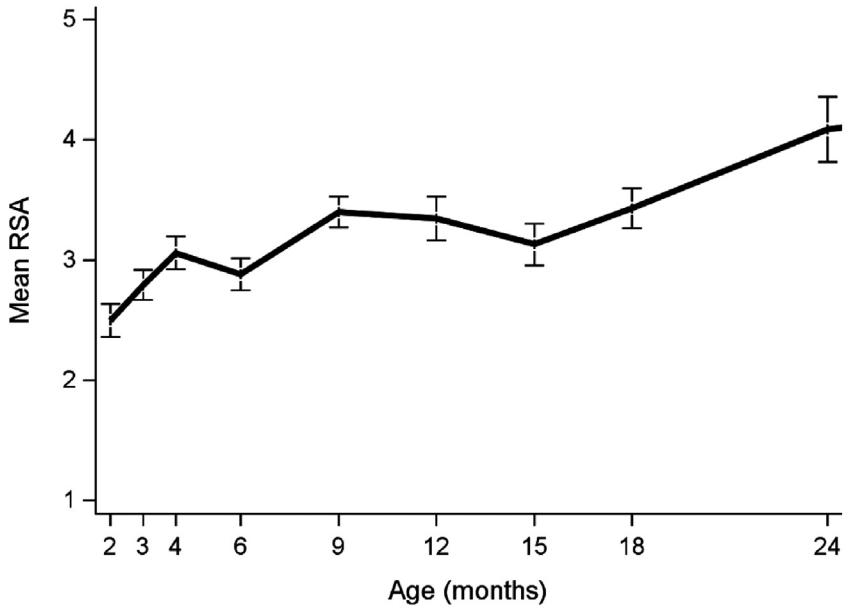


Fig.1 Trajectory of mean, resting RSA in typically developing infants from 2 to 24 months of age.

their infant for 2 minutes while maintaining a neutral expression. Because they were no longer face-to-face during this time, both the infant and caregiver were free to visually explore their environment, and the infant was able to move as they liked. After this 2-min period, the caregiver interacted with their infant again for 2 min. RSA was measured during each phase of this interaction, and RSA suppression was indexed as the difference between RSA during the “play” phase and RSA during the “ignore” phase. We found that RSA suppression during this task at 3–4 months positively associated with parent-reported social communication skills at 9 months. These findings suggest that appropriate parasympathetic responsivity to environmental demands, particularly those that are social in nature, may initiate developmental cascade that supports early social learning, which in turn fosters early social-communication development.

2.1 Autonomic dysfunction and autism in preterm infants

Decreased parasympathetic influence on the heart, reflected in lower HRV and RSA, can signal underlying ANS dysfunction (Thayer et al., 2009; Mathewson et al., 2014; Propper, 2012). Consistent with studies of children

and adults, and theoretical frameworks that describe ANS as a foundational regulatory function that is disrupted in ASD (Cheng et al., 2020; Patriquin et al., 2014; Sheinkopf et al., 2019), we have found differences in early ANS function and a high prevalence of ASD in infants born preterm. Preterm birth presents a unique context for studying ANS dysfunction, as these infants experience major challenges in early autonomic development. Substantial ANS maturation occurs in the second half of pregnancy, and preterm birth (born <37w GA) significantly disrupts the development of ANS, along with the overall functioning of the PT newborn (Mulkey & Plessis, 2018). The drastic transition to the extrauterine environment that occurs amidst ongoing ANS maturation is a clash that is compounded by exposure to medical stressors. As a result, autonomic maturation is impaired throughout the PT period, fails to normalize at term age equivalent (Fyfe et al., 2015), and may alter long-term temporal programming of ANS (Haraldsdottir et al., 2018). This early process initiates a developmental cascade that disrupts initial and subsequent physiological adaptations to the environment, and may affect the integration of autonomic control into later-developing, higher-level cortical processes important for social learning and cognitive development, such as attention. These early disruptions may be a critical factor in the emergence of ASD features, leading to heightened likelihood of ASD in very PT infants compared to the general population.

Building on the theoretical and empirical evidence linking ANS dysfunction to PT infants and to ASD, we conducted a prospective, longitudinal pilot study of very preterm infants (born < 32 weeks gestational age) in which we examined the predictive link between neonatal ANS dysfunction and later-developing ASD symptoms (Bradshaw et al., 2024). In this study, we collected continuous, multimodal measures of ANS dysfunction. ANS measures comprised minute-by-minute thermal gradients, defined as the difference in neonatal peripheral temperature and core temperature (Knobel-Dail et al., 2017), and hour-by-hour abnormal heart rate indices, measured using Heart Rate Characteristic (HRC) scores derived from HeRO Monitors, (Fairchild & Aschner, 2012). Heart Rate Characteristic Scores reflect an algorithm that considers beat-to-beat heart rate, heart rate variability, inter-beat interval (IBI) standard deviation, and IBI skewness (Fairchild & Aschner, 2012). Thermal gradients and HRCs were collected for 28 days after birth (>40,000 samples/infant) in the neonatal intensive care unit (NICU). Throughout the 28 days, ANS dysfunction was characterized by an average HRC score >1 and abnormal negative thermal gradients for about 20%

of the time (6 of 28 days). Following NICU discharge, comprehensive neurodevelopmental follow-up consisting of standardized measures of cognition, language, and motor skills were collected at adjusted ages 6, 9, and 12 months. Adjusted ages equate the neurological and physical development of preterm infants with that of full-term infants, whereas chronological age equates duration of time in the extrauterine environment with that of full-term infants. At 12 months adjusted age, assessments of social communication and early ASD symptoms were administered using the Systematic Observation Rating of Features of ASD (SORF; [Dow et al., 2017](#)).

In this small pilot study of very preterm infants who were retained in the study and followed to 12 months, we observed that the neonatal abnormal heart rate characteristics were strongly correlated with ASD symptoms at 12 months ($r = 0.81, p < .01$), as was birth gestational age, birth weight, and abnormal negative thermal gradients. A regression model including GA, BW, and HRC scores revealed a significant, unique predictive effect of abnormal heart rate indices on ASD outcomes (model $r = 0.86, p = .017$; HRC Score: $r = .72, p = .029$). Overall, these results show a surprisingly high association between ANS dysfunction in the neonatal period and ASD features at 1 year of age in very PT infants. This study provides compelling evidence for the role of ANS dysregulation and its cascading effects in the development of ASD. Whether this points to ANS as a catalyst for an ASD-specific developmental cascade, or reflects a developmental endpoint in an earlier cascade that serves as a more direct mechanistic pathway to ASD remains unknown, but highlights the need for further research into early autonomic mechanisms as predictors of ASD.



3. The role of attention as an early regulatory mechanism in ASD

Although atypical attention does not constitute a core feature of ASD, mounting evidence suggests impairments in attentional networks are pervasive and lifelong, constituting a central feature of ASD. Attentional disengagement is among the earliest reported features of ASD, particularly in EL infants ([Zwaigenbaum et al., 2005](#); [Elsabbagh et al., 2009](#)). Attention disengagement is the process of shifting attention from one stimulus to another and serves as a regulatory mechanism that modulates arousal. A bidirectional relationship between arousal and disengagement would involve effective arousal modulation facilitating attention disengagement, and failure to disengage leading to further dysregulation.

Our research demonstrates that atypical attention can be observed as early as three months of age in EL infants (Bradshaw et al., 2020) and infants who go on to develop ASD (Bradshaw et al., 2021). In these studies, we used a neonatal neurobehavioral exam to evaluate attention throughout the first five months of life. In the first study, we examined trajectories of visual and auditory attention to animate (faces and voices) and inanimate (objects) from 1 week to 5 months of age in EL infants and compared them to LL infants. We also tested associations between early attention and 12-month social-communication and cognitive outcomes.

Trajectories of attention across the first five months of life illustrated significant growth over time for both EL and LL infants. However, EL infants exhibited lower performance on the attention tasks at 2 and 3 months, particularly in visual attention tasks, compared to their LL peers. However, this difference was transient and by 4 and 5 months, their performance aligned with that of LL infants. Contrary to our expectations that social attention (i.e., animate attention tasks) would diverge most from that of LL infants, we found that attention to both animate and inanimate visual stimuli was lower during this time period, highlighting generalized vulnerabilities in attention during early development. At 12 months, early attention at 3 months was significantly associated with social-communication skills and, to a lesser extent, nonverbal cognitive skills.

In a follow-up study, we compared the same attention measures in infants who went on to develop ASD with those of typically developing infants (Bradshaw et al., 2021). Based on our earlier findings (Bradshaw et al., 2020), we focused on attention at 1, 2, and 3 months. Results showed that attention development differentiated infants with ASD from their TD peers. At 1 month, attention scores were comparable between groups, but by 2 and 3 months, infants with ASD exhibited significantly diminished attention. Further exploration revealed that this finding was most robust for visual, inanimate, attention (attention to objects). The difference in inanimate attention at 3 months in infants later diagnosed with ASD may reflect disruptions in the transition from reflexive to volitional control of attention during a critical period of neurodevelopment.

At birth, attention to visual stimuli is primarily subserved by subcortical mechanisms that enable reflexive orienting. This allows the youngest of neonates to orient to and track objects and faces. By 2 to 3 months, however, cortical systems, particularly those in the parietal and prefrontal regions, begin to mediate volitional attention and attentional shifting. This transition supports the development of more complex behaviors, such as

joint engagement with objects and caregivers, which are foundational for social and communication skills. Infants with ASD may experience delays or disruptions in this cortical-driven process, resulting in diminished ability to orient to and sustain attention on inanimate stimuli. Moreover, orienting requires visual disengagement from other salient stimuli and visual tracking requires motor control and movement of head and neck that aligns with eye movements. Both attention disengagement and motor challenges, particularly head, neck, and trunk control (Bradshaw et al., 2023), are observed in very early development in ASD. These difficulties with attention disengagement from social or environmental stimuli, as well as impaired coordination between visual and motor systems, may also explain the attention differences observed in these studies. Together, these results emphasize the importance of attentional processes during the 2- to 3-month window, possibly the result of underlying neurodevelopmental or regulatory vulnerabilities, which may shape developmental pathways later observed in ASD.



4. The coupling of ANS and attention: autonomic regulation of attention

Attention and parasympathetic activation both serve as foundational regulatory functions throughout the lifespan. In infancy, the coupling of autonomic regulation and attention is a cornerstone of an adaptive engagement system. The ANS modulates attention by regulating heart rate responses during information processing, particularly through parasympathetic activity. A well-functioning ANS enables neonates to override metabolic demands and generate physiological responses, such as heart rate deceleration, which support sustained attention and enhance stimulus processing (Lester et al., 1996; Richards, 2004; Richards & Turner, 2001).

Parasympathetic-driven attention responses underpin three distinct phases of attention—attention orienting, sustained attention, and attention termination—each linked to cognitive and learning processes (Richards, 2010; Richards, 2004). Among these, sustained attention, plays a particularly crucial role in cognitive engagement. Characterized by prolonged look duration and heart rate deceleration, sustained attention facilitates prioritization of relevant stimuli while reducing distractibility, resulting in selective information processing and recognition memory (Richards & Hunter, 1998; Reynolds & Richards, 2005).

Heart rate measures of sustained attention represent a noninvasive method for understanding cognitive and neurological processes that underly

sustained attention. During sustained attention, the heart rate slows and remains decelerated, signaling heightened cognitive engagement and information processing. As sustained attention shifts to the third phase, attention termination, heart rate returns to baseline, reflecting diminished arousal and increased distractibility (Richards, 2004). Sustained heart rate deceleration reflects the activation of arousal systems, crucial for maintaining attention, while the return to baseline during termination marks disengagement (Richards & Casey, 1992). Heart rate-defined sustained attention (HRDSA) can be indexed with duration of the heart rate deceleration, and the magnitude of the heart rate change, or the depth of deceleration. HRDSA is said to offer a more precise measure of information processing than looking behavior alone (Colombo, 2001; Reynolds et al., 2011; Xie & Richards, 2016). This sequence highlights the dynamic role of parasympathetic activity in regulating attention and underscores the importance of HRDSA as a measure of meaningful attention and engagement during infancy.

4.1 Developmental course of autonomic regulation of attention

Attention undergoes significant changes during the first two years of life, shaped by the maturation of attention networks. Early in infancy, posterior attention systems support visual orienting and disengagement, while anterior attention systems involved in executive control emerge after six months. These neurodevelopmental milestones contribute to varied trajectories in attention metrics. For instance, look duration, which indicates visual learning and encoding, follows a triphasic trajectory: increasing during the first two months, decreasing between three and six months, and rising again after six months (Colombo, 2001; Richards & Anderson, 2004).

To further investigate the dynamic unfolding of these regulatory processes, and to comprehensively document the development of attention in context, we examined age- and context-dependent differences in looking behavior and physiologically-regulated sustained attention. In a two-year longitudinal study, we mapped trajectories of attention in over 50 infants from 1 to 24 months of age. To understand the effect of stimulus on attention patterns, we included four distinct stimulus conditions—social videos, nonsocial videos, social interactions, and nonsocial interactions—across multiple attention metrics that included overall looking time, mean look duration, percent time in HRDSA, and heart rate deceleration during HRDSA episodes.

Results of this study showed that during the first six months, overall looking time increased across all conditions, with screen-based tasks eliciting more looking than live interactions. However, contrary to expectations, a drastic decline in attention to the caregiver's face during the social interaction was observed between 4–6 months and was lower than all other stimuli by 6 months. The decline in face-looking coincides with developmental gains in motor skills and executive control of attention, which facilitates a broader, more dynamic exploration of the environment. The emergence of these skills is also closely tied to the maturation of cortical control of attention, particularly the development of the anterior attention network. This network underpins voluntary attention shifting and disengagement, processes that are crucial for adapting to and learning from a complex, multimodal environment. Importantly, our data showed that HRDSA during face-looking remained stable or increased slightly during this period, suggesting that while infants may look at caregivers' faces less frequently between four and six months, these looks reflect stable cognitive engagement and efficient processing. This highlights the value of HRDSA as a measure of meaningful attention.

In the second year of life (9–24 months), attention measures become more stable across all task conditions, with videos (both social and nonsocial) eliciting more sustained attention than live toy interactions. This may be due to the perceptual saliency and cognitively engaging content of videos compared to the real-world demands of interactive tasks, which require the infant to initiate the interaction and engage motor and cognitive resources for sustained engagement. Social videos elicited longer look durations compared to nonsocial videos during this period, suggesting an enhanced attention to social stimuli as infants' communication and language skills develop.

Findings from this study highlight the importance of considering both context and measurement method in studies of infant attention. Our results suggest that behavioral measures—percent looking and look duration—capture general behavioral trends, while HRDSA and heart rate deceleration provide deeper insights into cognitive engagement. The divergence between these attention measures demonstrates that reduced looking time does not necessarily indicate diminished attentional engagement, especially in dynamic, interactive contexts. This underscores the need for multi-method approaches to studying infant attention, particularly for identifying early markers of atypical development. Overall, this study establishes a multimethod-informed trajectory of typical infant attention over the first

two years of life, providing a reference point for comparisons with other populations, such as EL infants or those with ASD.

4.2 Autonomic regulation of attention and ASD: a developmental framework

The evidence is clear that physiological measures of attention, reflecting ANS's regulation of attention, are related to cognition and learning. Early disruptions in autonomic regulation of attention then may have cascading consequences for interaction, learning, and development (Feldman, 2006; Porges & Furman, 2011; Bradshaw et al., 2022). Based on current literature and our preliminary data, we propose a developmental framework that highlights the role of ANS dysregulation in attention very early in development and has cascading effects on learning and development. In this framework, global ANS function, which supports neonatal arousal and reactivity, appears intact at birth. Arousal and reactivity during this period are primarily driven by sympathetic activation, allowing newborns to respond to environmental demands, such as crying in response to internal states like hunger and pain. However, as parasympathetic regulation matures during infancy, it has increasing influence over the heart and becomes critical for maintaining a calm state and modulating attention. Adaptive modulation of arousal and attention then supports sustained cognitive engagement and readiness for social interaction. In infants who develop ASD, this transition to effective parasympathetic regulation may be disrupted.

This disruption may underly observed attention differences in the 2–3 month period (Bradshaw et al., 2020; Bradshaw et al., 2021), when parasympathetic modulation of attention becomes increasingly necessary for adaptive functioning. Ineffective parasympathetic modulation interferes with the ability to sustain attention to relevant stimuli and flexibly shift attention to novel, salient stimuli. This impairment may diminish the quality and frequency of learning experiences, particularly those that involve social and communicative interactions. Over time, the resulting cascade may contribute to the emergence of core ASD features, including difficulties in social communication and behavioral flexibility.

Preliminary data that support this framework include trajectories of heart-defined sustained attention during naturalistic parent–infant interactions from 1–4 months across EL, LL, and PT infants. These trajectories are characterized by a significant increase in heart rate (HR) deceleration during sustained attention and point to a dampened HR deceleration that

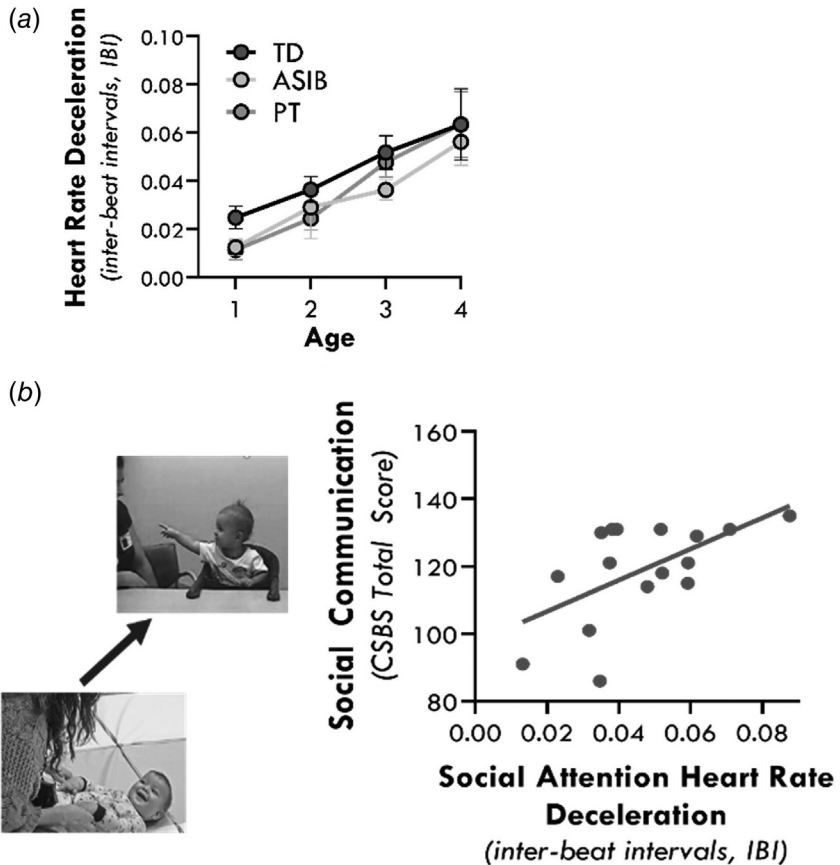


Fig. 2 a. Developmental trajectory of heart rate deceleration within periods of heart rate-defined sustained attention (HRDSA) during infant-caregiver interactions across ASD, TD, and PT infants. **b.** Association between deceleration within periods of HRDSA measured during infant-caregiver interactions at 3-4 months and social-communication skills measured at 9 months.

is developmentally transient for PTs, but persistent for EL infants (Fig 2A; (Bradshaw et al., 2022b)). Moreover, we have found that HR deceleration during sustained attention to caregivers at 3 months is associated with early-emerging symptoms of ASD (Bradshaw and Abney, 2021) (Fig. 2B).

In a preliminary investigation and an extension of our previous work (Bradshaw et al., 2024), we examined physiological measures of attention to social and nonsocial videos across EL and LL infants from 1 to 24 months of age. While LL infants demonstrated an emerging distinction between social and nonsocial attention based on both behavioral and physiological

measures in the second year of life, with greater attention to social stimuli, this differentiation was absent in EL infants. Specifically, for EL infants, the proportion of time spent looking to the screen and the proportion of looking classified as HRDSA were equivalent across both types of stimuli. This was not driven by diminished attention to both social and nonsocial stimuli, but rather high levels of both social and nonsocial attention. Trajectories of social attention between EL and LL infants did not differ. These findings suggest that from 12 to 24 months, LL infants exhibit enhanced attention to social stimuli or attenuated attention to nonsocial stimuli. In contrast, EL infants engage in equally high levels of attention to both social and nonsocial stimuli, reflecting a lack of differentiation in attentional prioritization that may underlie or reflect early disruptions in social engagement.



5. ANS and attention: cascading consequences in ASD

In ASD, the transition to effective parasympathetic modulation of ANS in the first year of life emerges as a critical developmental inflection point. ANS dysregulation during this period may limit the capacity for sustained attention and adaptive engagement. This framework has yet to be fully tested, but there are at least two alternative consequences to a dysregulated autonomic-attention system. First, instead of ANS dysregulation cascading into a less efficient attention control system, compensatory mechanisms may lead to the need for increased parasympathetic activity, evidenced by deeper heart rate decelerations and higher RSA, needed to achieve similar levels of regulation and attention as neurotypical infants. And second, there may be a decoupling of ANS regulation and attention mechanisms in ASD, meaning that other systems are employed to regulate attention and that parasympathetic influence over the heart may not necessarily serve to regulate attention or other processes. This may lead to the deployment of other systems and mechanisms to facilitate regulation. These other systems may not only be less effective, but may take away from resources allocated to other neurodevelopmental processes.

Taken together, these findings emphasize the importance of examining parasympathetic function and autonomically regulated attention as both a developmental resource and a potential early marker of ASD. By integrating behavioral and physiological measures of attention, such as HRDSA, future research can refine models of neurodevelopment and identify early pathways leading to ASD. This approach offers promise for earlier detection and intervention strategies that target attention and autonomic regulation to

mitigate developmental disruptions and promote adaptive learning and interaction.



6. Multimodal methods in naturalistic contexts: an approach to understanding heterogeneity in ASD

The phenotypic manifestation of ASD is strikingly variable across individuals. This heterogeneity is apparent in early developmental pathways as well. While some infants exhibit clear trajectories toward an ASD diagnosis, others show more variable patterns, including transient symptoms, or delayed onset of early ASD markers and behaviors (Landa et al., 2007). Through a developmental cascades framework, we understand environmental contexts play an important role in developmental trajectories and outcomes. The effect of environmental contexts on developmental trajectories of ASD is understudied, but interactions between biological and experiential factors likely influence the observed heterogeneity in outcomes. While statistical models can, and often do, control for context variables like socioeconomic status (SES) or early intervention history, these adjustments may not fully account for the nuanced variability observed in naturalistic social interactions or home environments. As an example, differences in motor actions and possibly motor skill development in infancy can be influenced by the home environment, even subtle features like furniture arrangement or flooring textures. For instance, a bedroom with limited furniture at the height level of an infant might lead to fewer opportunities for cruising behavior, an important bridge milestone to independent walking. Alternatively, an infant in this environment may benefit from enriched social interactions as caregivers provide support for the infant to cruise while they practice walking. The child's behavior is therefore a result of the interaction between the child's own abilities and the environmental opportunity to develop and express these abilities. Similarly, infant behavior and physiological regulation, measured with RSA, have been found to be modulated by maternal sensitivity (Girod et al., 2024), highlighting how physiological and behavioral differences emerge only under specific contextual conditions. Together, these examples highlight the need to study early markers situated within an individual's environment, familial context, behavior, and biology. These nuanced, interconnected dynamics set the stage for examining how specific physiological and attentional mechanisms interact with environmental contexts to uncover specific developmental pathways of ASD.

6.1 Naturalistic studies of attention and ANS

Thus far, we have discussed studies of physiological regulation and attention that occur in largely controlled, lab-based environments. While these studies provide valuable insights into the mechanisms underlying early differences in infants who develop ASD, they often fail to capture the dynamic and multifaceted nature of real-world infant experiences. In everyday life, infants are immersed in rich, complex environments where attention is constantly influenced by interactions with caregivers, fluctuating contextual demands, and evolving developmental capacities. These complex factors both shape and are shaped by the infant's ongoing development. Naturalistic settings present challenges for researchers aiming to measure attention and other neurodevelopmental processes in a way that reflects their real-world functionality while maintaining methodological rigor.

To address this gap, we have leveraged specialized approaches that bridge two highly valued features of developmental science: experimental control and naturalistic contexts. Advanced tools like head-mounted eye tracking (HMET) capture infants' moment-to-moment visual engagement within everyday social environments. This method allows us to observe how infants allocate their attention to social and nonsocial features of their environment as they interact with their caregivers and surroundings. By applying this approach in infant sibling studies of ASD, we aim to uncover nuanced patterns of attention that are both influenced by and predictive of early developmental trajectories. These efforts provide a more comprehensive picture of how attention unfolds in the dynamic, real-world contexts shaped by social, motor, and cognitive development.

6.2 Feasibility of head-mounted eye tracking in very young infants

Infant social attention develops within an intricate, embodied context shaped by dynamic interactions with caregivers, objects, and the broader environment. Traditional, lab-based approaches fail to capture the real-time, bidirectional processes through which caregivers scaffold infant behavior and infants explore their surroundings. HMET technology offers an innovative lens to study infant social attention within naturalistic and even home-based contexts. Unlike traditional screen-based eye-tracking methods that constrain participants to fixed positions and controlled environments, HMET provides the unique ability to capture real-time gaze patterns as individuals move, interact, and engage with their surroundings. By

integrating lightweight cameras mounted on a headband or cap, researchers record both the wearer's gaze and their field of view, enabling precise mapping of attention within three-dimensional, everyday environments. This approach is particularly valuable for studying infants, whose behaviors and interactions unfold in dynamic and unpredictable ways that cannot be fully captured in static laboratory settings (Bradshaw et al., 2023). Recording gaze behavior during naturalistic interactions facilitates the study of how attention is dynamically shaped by factors like body posture, movement, and environmental affordances. Moreover, HMET enables researchers to capture moment-to-moment coordination between individuals, such as how a caregiver's gaze and actions influence an infant's attention or how joint attention is established during shared activities. These capabilities make HMET an essential tool for understanding the embodied and interactive nature of attention, particularly in populations with atypical development, such as children with ASD, where traditional paradigms may not fully capture the nuances of their real-world social behavior.

The first study to employ HMET with children diagnosed with ASD offered surprising results that challenge longstanding models of social attention deficits in ASD (Yurkovic-Harding et al., 2022). Contrary to the extensive literature emphasizing social attention deficits, in particular gaze patterns to faces, and impaired joint attention skills in children with ASD, this research revealed that children with ASD show no significant differences in real-world attention to faces during free play with a caregiver compared to their typically developing peers. Both TD children and those with ASD spent minimal time looking at their caregiver's face during naturalistic play, with face-looking accounting for less than 2% of the interaction time. These results are particularly striking considering the vast research landscape illustrating social attention deficits in ASD that has guided our understanding and core models of ASD etiology.

In addition, the study examined joint attention, or coordinated visual attention (CVA), between children and caregivers. CVA is defined by moments of child and caregiver simultaneous gaze to the same object. Again, both groups demonstrated comparable rates of CVA, predominantly via alternative pathways, such as manual interactions with objects. This observation also runs counter to prevailing notions that gaze to face is the primary conduit for establishing shared attention, or CVA. In the early ASD literature, joint attention, which involves CVA with a social partner for the explicit purpose of sharing attention to the same object or event, is predominantly characterized by discrete instances of child behavior,

including point- and gaze-following (child response to joint attention), or distal pointing and/or two-point and three-point gaze shifts between faces and objects (child initiation of joint attention). These cues represent the primary pathways to joint attention. While deficits in these pathways are well-documented and clinically significant (Adamson et al., 2009; Dawson et al., 2004; Leekam & Ramsden, 2006; Naber et al., 2007; Nyström et al., 2019; Warreyn et al., 2007), Yurkovic-Harding et al. (2022) suggests they may represent reliable markers of ASD rather than causal mechanisms underlying the disorder.

The study's context—a naturalistic, free-flowing play environment—exemplifies the critical importance of context and advanced tools that effectively capture the breadth and impact of this context. Unlike the constrained settings of traditional laboratory-based paradigms, the use of HMET in this study captures the complexity of real-world interactions, during which infants and caregivers co-construct shared attention through dynamic, multimodal exchanges. By highlighting the potential for alternative pathways to support social and communicative development, this research opens new avenues for understanding developmental trajectories in ASD.

Leveraging the strengths of HMET in naturalistic contexts, we have explored how this tool can be applied in the earliest stages of infancy to examine embodied attention in everyday environments. Existing HMET research has largely focused on toddlers and children, leaving a gap in how infants, particularly those as young as four months old, coordinate attention with their caregivers and surroundings. In a longitudinal study of infants at low and elevated familial likelihood for ASD, we evaluated the feasibility, acceptability, and data quality of using HMET with four- and eight-month-old infants (Bradshaw et al., 2023). These age points represent developmental periods of motoric transitions and achievement of key milestones, that is, reaching and crawling. The eight-month study age point also represents a formative developmental period just preceding the beginning emergence of social and communicative behaviors. By situating these investigations in the home environment, where infants engage in the vast majority of interactions with their caregivers, this study pushes the boundaries of what can be achieved with wearable eye-tracking technology. As such, a key goal of the study was to evaluate whether HMET could be successfully implemented in the home with younger infants who lack independent head control—a population previously understudied due to these technical challenges—and to assess the acceptability of these procedures for caregivers. Additionally, the study aimed to explore early developmental changes in infant gaze behavior,

particularly shifts in attention between caregivers and objects, and how these patterns might vary by age or ASD likelihood.

The application of HMET to infants as young as 4 months in the home environment proved to be highly feasible. A custom headband was designed for 4-month-olds that prioritized comfort and camera stability, as even slight shifts can have significant impacts on data quality. All infants tolerated the placement of the headset, and the data collected met established criteria for accuracy and precision, with only minimal data loss across sessions. Importantly, caregivers reported high levels of acceptability, with most indicating that the experimental procedures did not disrupt their infant's natural behavior. The retention rate from the 4-month to 8-month visit was also high, reflecting strong engagement from participating families. Together, these findings suggest that HMET, when paired with appropriate equipment and methodological adjustments, can be successfully adapted for use in a home setting and with young, possibly neurodivergent, infants—even those as young as four months of age.

6.3 Gaze patterns in naturalistic contexts in young infants at elevated likelihood for ASD

Preliminary analyses of gaze behavior reveal intriguing developmental trends. At 4 months, infants spent significantly more time looking at their caregiver's face compared to 8-month-olds, whose gaze was more frequently directed toward objects in the play environment. This pattern reflects developmental shifts in social and attentional priorities, as younger infants rely heavily on face-to-face interactions while older infants begin to explore their surroundings more independently. Notably, these trends align with existing literature demonstrating a decline in face-looking and an increase in object-focused attention across the first year of life (e.g., [Bradshaw et al., 2024](#)).

In addition, these results extend those of [Yurkovic-Harding et al. \(2022\)](#) showing that despite the availability of caregiver faces for the vast majority of the time during the 4-month-old infant-caregiver interactions (76%), their gaze to face was strikingly low (16%). At 8 months, availability of the caregiver's face was dropped significantly (only 17% of the interaction) and proportion of time infant's gazed to the caregiver's face was only 5%. These results seem to challenge long-standing assumptions driven by evidence of infants' attentional bias toward faces in lab-based settings (e.g., [Leppänen, 2016](#); [Peltola et al., 2018](#); [Reynolds & Roth, 2018](#)). While faces are often emphasized as highly salient in controlled settings, naturalistic data reveal more nuanced patterns of social attention, shaped by real-world

complexities. The home environment offers a rich, dynamic backdrop for studying how infants and caregivers co-construct interactions, with caregivers often scaffolding attention through behaviors like naming, touching, and positioning objects. By capturing these behaviors *in situ*, HMET allows researchers to examine not only what infants attend to but also how their attention is guided and supported by their social partners.

We have also explored CVA and the pathways into CVA among 8-month-old EL and LL infants. Analyses identified three key differences between the groups. We found that look durations to toys were greater for EL infants and bouts of CVA were fewer in EL infants compared to LL infants. This aligns with deficits in attention shifting and attention disengagement that is observed in infants and toddlers who later develop ASD. Additional analyses are needed to thoroughly interpret this preliminary finding, but it is likely that EL infants attend to the same toy for longer periods, and their caregiver follows their attention, leading to fewer CVA bouts. In addition, we see that EL infants' CVA episodes are more often infant-initiated, with 66% of CVA bouts in the EL group being infant-led. That is, the CVA bout began with the infant directing their gaze toward a toy, followed by the parent coordinating their attention to the same toy. This proportion was significantly higher than that observed in LL infants, whose CVA episodes were more balanced between parent-led (47%) and infant-led (53%) pathways.

Together, these studies offer new perspectives on how lab-based mechanistic studies of infant attention may translate to and inform studies of attention allocation in real-world settings—the very contexts that shape and support development and learning. Leveraging HMET and other technologies to study embodied attention and natural behavior, we can further understand attention, not only as a set of discrete behaviors, (e.g., attention disengagement, attentional social preference) but as an emergent property of dynamic interactions, offering new pathways to understanding how developmental trajectories take shape in both typical and atypical populations.

6.4 A synchronized system for studying physiological regulation of attention in naturalistic settings

HMET is not the only advanced tool that can be made portable and implemented in the home to study developmental trends in infant development. In our longitudinal studies of EL and LL autonomic regulation of attention,

we have used home-based data collection to study behavior in naturalistic contexts and to address barriers to access for research participants. Our team has developed the Lab2Home (L2H) system – a tailored, portable, and scalable apparatus designed to bring rigorous laboratory-quality data collection directly into participants' homes. The L2H system is tailored for studies requiring synchronized data streams from multiple video angles and concurrent heart rate monitoring for both parent and child. Synchronization, a cornerstone of L2H, is achieved using a combination of custom-written and open-source software with redundant methods to ensure millisecond precision. Devices utilize a shared computer clock for alignment of time-stamped data, while a custom-built microcontroller triggers an LED array recorded by all cameras and logged into the experimental system for added accuracy. Additionally, visual time displays are embedded into recorded video streams to align seamlessly with heart rate data. These measures ensure that all data modalities remain synchronized throughout the procedure, even in diverse and variable home environments.

L2H is compact, lightweight, and designed for rapid setup and break-down, reducing logistical overhead during testing. The system's hardware includes three high-definition (1080p, 60fps) USB-C cameras mounted on adjustable tripods to accommodate a range of home layouts, from spacious living rooms to confined apartments. High-capacity portable power banks power the system for over 10 hours, allowing uninterrupted operation even in homes without access to reliable power. Heart rate data collection is facilitated by two CamnTech Actiheart-5 wireless monitors—one for the parent and one for the child—while a small-profile condenser microphone with preamp ensures high-quality audio recording and strategic placement. The central microcontroller, an Arduino UNO WiFi REV2 outfitted with a multicolor LED array, coordinates the system's operations seamlessly.

The L2H apparatus not only facilitates the collection of rich, high-quality data but also reduces the systemic barriers that often limit the participation of historically underrepresented populations in research. By enabling data collection in familiar home environments, L2H supports studies that aim to understand developmental processes in the very contexts in which they unfold.



7. In-Home studies as a representative and inclusive approach to science

Naturalistic data collection, particularly in the home, offers a unique window into infant behavior and development in real-world contexts,

affording the possibility of more ecologically valid data while maintaining high standards of rigor. Moreover, the cost of travel for participants, in terms of time, vehicle mileage, and gas expenses, decreases accessibility of research participation for families from diverse sociodemographic backgrounds, even when monetary compensation is provided. Inclusivity and representation are incredibly important to confirm our science to be representative of and generalizable to the broader population at large.

To evaluate the experimental and ecological validity of in-home data collection, we administered parallel experimental and naturalistic tasks longitudinally to two groups of infants who participated in in-lab study visits ($n = 29$, 63%) or in-home study visits ($n = 17$, 37%). Infants were seen from 2–24 months of age, resulting in a total of 167 lab visits and 106 home visits. Following data collection, caregivers rated their child's observed emotional and interactive states. Caregiver report of emotional reactivity during experimental and clinical procedures significantly differed by context ($\chi^2(2) = 12.64$, $p = 0.002$) with lab-based sessions yielding more parent-reported atypically negative emotional reactivity ($n = 23$; 13.8%) compared to home-based sessions ($n = 4$; 3.8%). Ratings of interest and attention during the tasks did not significantly differ by setting ($\chi^2(2) = 2.76$, $p = 0.252$). These findings suggest that the controlled lab environment may contribute to more atypical behavior, that is, negative emotional reactivity, suggesting more typical behavior in the home environment and possibly more representative and valid data. Experimental data from variables of interest was also comparable across home vs. lab contexts. The percentage of valid, usable behavioral and electrocardiogram (ECG) data was not significantly different ($p = 0.73$, $p = 0.89$, respectively).

A home-based approach may also increase inclusivity and accessibility of participant cohorts. Traditional lab-based studies often fail to engage families living in rural areas or from underrepresented socioeconomic groups, such as Black, Indigenous, and people of color (BIPOC). By bringing research into the home, we expand the demographic reach of participants, paving the way for a representative, enriched dataset that includes diverse populations. In our analysis of a separate cohort of participants who opted to complete home vs. lab study visits (see [Table 1](#)), household income significantly differed ($\chi^2(1) = 8.99$, $p = 0.003$), with 40.7% of home-enrolled participants earning \$60,000 or less annually compared to 11.1% of lab-enrolled participants—well below the state average household income (South Carolina; \$92,833). As expected, the average distance from the participant's home to the lab was significantly higher for the home-enrolled participants ($t(131.5) =$

Table 1 Participant demographics of participants across lab vs. home-based data collection.

	Lab enrolled, N = 40	Home enrolled, N = 103	Test statistic (p value) ^b
Child race			
White-non-Hispanic	26 (65%)	51 (49.5%)	0.257
Black or African American	9 (22.5%)	33 (32%)	
More than one race	3 (7.5%)	12 (11.7%)	
White-Hispanic	1 (2.5%)	4 (3.9%)	
Asian	0 (0%)	3 (2.9%)	
Other/Unknown	1 (2.5%)	0 (0%)	
Child race ^a			
Black, indigenous, and persons of color (BIPOC)	14 (35%)	52 (50.5%)	0.139
White	26 (65%)	51 (49.5%)	
Reported Primary Caregiver			
Mother	37 (92.5%)	102 (99%)	0.119
Father	3 (7.5%)	1 (1%)	
Primary Caregiver Education ^a			
High School/GED/some college	12 (30%)	47 (45.6%)	0.129
Bachelor/Graduate/Professional degree	28 (70%)	56 (54.4%)	
Not reported	2	0	
Annual Household Income*			
<US\$20,00–\$60,000	4 (11.1%)	37 (40.7%)	0.003
US\$60,001 or higher	32 (88.9%)	54 (59.3%)	
Not sure or prefer not to answer	4	12	
Distance from the Lab			
Average roundtrip mileage (SD)	23 (22)	73.5 (76.6)	<0.001

^aDichotomized groups and chi squared analyses.^bBold values indicate statistical significance at the $p < .05$ level.

-6.0, $p < 0.001$). Results from these analyses suggest that distance to the lab stands as one of the barriers to lab-based participation and that offering in-home study visits increases accessibility of research participation for lower-resourced families.

Confirming that findings from controlled, lab-based experiments completed in the home are comparable to those completed in the lab, while also demonstrating differences in infant behavior and sociodemographic factors across the two contexts, underscores the feasibility and impact of in-home study visits. To continue this work, we have developed a sophisticated method using portable, synchronized technologies that can effectively

capture synchronous behavioral and physiological data without disrupting the familial environment. By situating research in participants' homes, with advanced technology explicitly designed for our study procedures, we aim to gain a deeper understanding of how early ASD markers manifest in naturalistic settings and reduce barriers for families from diverse demographic backgrounds.



8. Conclusions

The science of ASD in infancy is at an exciting stage, offering new opportunities to elucidate the mechanisms and developmental cascades underlying atypical neurodevelopment in real-world contexts. The naturalistic, behavioral, and non-invasive nature of this work elevates its potential for rapid translation to novel and, most importantly, actionable early detection and intervention strategies. By integrating a developmental cascades approach with prospective, longitudinal designs, innovative methodologies, and home-based data collection, we can illuminate new pathways that characterize early development of ASD. Grounded in both mechanistic insights and real-world applicability, we can understand how autonomic regulation and attentional processes shape developmental outcomes.

Prioritizing inclusivity and ecological validity through home-based data collection not only broadens the scope of ASD research but also ensures its relevance to the lived experiences of diverse families. This multifaceted approach not only advances scientific understanding but also lays the groundwork for interventions that resonate with the complexities of infant development and with the cultural context of families traditionally under-represented in research. As ASD research continues to evolve, bridging theory, empirical findings, and methodological innovation will remain critical to transform outcomes for children and their families.

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